

Troubleshooting Guide: Nitrogen Generation Package (PSA) – Declining Product Purity

Category: Utilities / Gas Generation / Cyclic Adsorption **Unit:** Pressure Swing Adsorption (PSA) Nitrogen Generation Package **Tools:** Cycle-by-cycle product purity trending, adsorber bed pressure profile review **Fluid Package:** Not applicable in the VLE sense — PSA performance is governed by adsorption kinetics/selectivity, similar in category to the molecular sieve dehydration case (Case Study 10), not phase-equilibrium thermodynamics **Symptom:** Product nitrogen purity gradually declining below the required spec for its blanketing/inerting service, despite the PSA package running its normal automated cycle

Note on case type: Like Case Study 10 (molecular sieve dehydration), this is a **cyclic adsorption performance** investigation rather than a VLE-based diagnosis.

1. Quick Reference

Item	Detail
Reported symptom	Product nitrogen purity gradually declining below spec despite the PSA package running its normal automated cycle
Initially unclear	Whether this was carbon molecular sieve (CMS) media degradation (end-of-life) or a cycle-timing/valve performance issue preventing the beds from completing their intended adsorption/regeneration steps properly
Actual root cause	A slowly leaking cycle valve was allowing partial cross-flow between beds during what should have been an isolated step, effectively shortening the usable adsorption time and allowing oxygen breakthrough earlier than the cycle timer assumed
Fix	Repaired/replaced the leaking cycle valve; verified full isolation between beds during each cycle step
Diagnostic signal	Bed pressure profiles during cycle steps that should show a fully isolated bed instead showed a small but consistent pressure interaction with the adjacent bed, inconsistent with a properly sealed cycle valve
Prevention	Routine cycle valve leak/seat testing; per-cycle bed pressure profile review as a standing diagnostic, not just overall product purity monitoring

2. Symptom

- **Product nitrogen purity gradually declined**, falling below the specification required for its blanketing/inerting service.
- The PSA package continued to **run its normal automated cycle** — no obvious alarm or cycle fault was indicated by the control system.

3. Why This Wasn't Assumed to Be Simple CMS Media End-of-Life

Declining PSA product purity is often attributed to **carbon molecular sieve (CMS) media degradation** — a normal end-of-life process for this consumable, addressed by media replacement. Before committing to that (costly) conclusion, it was worth checking whether the **cycle itself** was executing correctly, since a valve or cycle-timing fault can produce the same purity-decline symptom as genuine media degradation, but with a very different (and cheaper) fix.

4. Diagnostic Approach

Step 1 — Review product purity trend for a run-time-correlated pattern

Purity decline was reviewed against cumulative operating time/cycles, checking for a smooth, aging-consistent decline pattern (as would be expected from genuine CMS degradation).

Step 2 — Review per-cycle bed pressure profiles

Rather than relying on purity trend alone, individual **bed pressure profiles** during each step of the automated cycle were reviewed in detail — specifically checking whether each bed was achieving true pressure isolation during the steps where isolation is required (e.g., during the other bed's regeneration/blowdown).

Finding: Bed pressure profiles showed a **small but consistent pressure interaction with the adjacent bed** during steps that should have shown full isolation — inconsistent with a properly sealed cycle valve.

Step 3 — Investigate the specific cycle valve responsible for that isolation step

The specific valve responsible for isolating the two beds during that cycle step was inspected and found to have a **slow internal leak**, allowing partial cross-flow between beds rather than the clean isolation the cycle design assumes.

Step 4 — Connect the leak to reduced effective adsorption time

With partial cross-flow occurring during what should be an isolated adsorption step, the **effective, usable adsorption time** for the bed currently producing nitrogen was being shortened — allowing oxygen to break through earlier in the cycle than the fixed cycle timer accounted for, since the timer assumed full isolation and a clean cycle.

Quantitative Basis

- Product spec for blanketing service: $\geq 99.5\% \text{ N}_2$ ($\leq 500 \text{ ppm O}_2$). Baseline: $99.7\% \text{ N}_2$ (300 ppm O_2). Declined to **$98.9\% \text{ N}_2$ ($11,000 \text{ ppm O}_2$)** — well below spec.
- 2-bed PSA, 60-second total cycle, 55 seconds nominal adsorption time per bed. During the adjacent bed's blowdown/purge step, the producing bed's pressure should hold steady at $100 \pm 1 \text{ psig}$; trend data showed it dipping to **94 psig** during that step — a cross-flow signature.

- Bench leak test on the isolation valve for that step measured **8 SCFH seat leakage, against a ≤ 0.5 SCFH allowable spec** — a 16x exceedance.
- Estimated effective usable adsorption time, accounting for the cross-flow loss, was **≈ 38 seconds instead of the nominal 55 — a 31% reduction**, consistent with oxygen breaking through well before the fixed 55-second cycle timer expected it.

5. Root Cause

A slowly leaking cycle valve was allowing partial cross-flow between adsorber beds during a step that should have been fully isolated, shortening the effective usable adsorption time for the producing bed and causing oxygen breakthrough earlier than the fixed automated cycle timer assumed — reducing product nitrogen purity even though CMS media itself had not reached genuine end-of-life.

6. Corrective Action

1. **Repaired/replaced the leaking cycle valve.**
2. **Verified full isolation between beds** during the relevant cycle step following the repair.

7. Verification

- Post-repair, bed pressure held steady at **99-100 psig** through the previously-affected isolation step, with no further dip.
- Product O₂ content dropped to **280 ppm**, giving **99.72% N₂ purity** — better than the original baseline.
- Held stable across **$\sim 43,200$ cycles (30 days of continuous operation)** with no recurrence of the decline.

8. Prevention / Long-Term Fix

- Established **routine cycle valve leak/seat testing**, since a slow internal leak like this can develop gradually and not trigger any cycle-fault alarm on its own.
- Added **per-cycle bed pressure profile review as a standing diagnostic**, rather than relying solely on overall product purity monitoring, which only reveals a problem after it has already affected product quality.

9. General Troubleshooting Checklist (Reusable)

- ☐ Review product purity decline against cumulative run-time/cycles for a pattern consistent with normal media aging
- ☐ Before assuming CMS/adsorbent media end-of-life, review **per-cycle bed pressure profiles** in detail, not just the overall purity trend
- ☐ Check specifically for pressure interaction between beds during steps that should show full isolation
- ☐ If found, identify and inspect the specific cycle valve responsible for isolation during that step
- ☐ Repair/replace a leaking cycle valve and verify full isolation is restored
- ☐ Confirm purity recovery following the valve repair before concluding media replacement is unnecessary

- ❑ Establish routine cycle valve leak/seat testing and per-cycle bed pressure profile review as standing diagnostics, since a slow internal leak can develop without triggering a standard cycle-fault alarm

10. Key Takeaway

Declining PSA product purity doesn't automatically mean the adsorbent media has reached end-of-life — a leaking cycle valve can produce the exact same symptom by quietly shortening the effective adsorption time each cycle, well before the fixed cycle timer would expect oxygen breakthrough. Reviewing per-cycle bed pressure profiles for proper isolation is a much cheaper and more specific diagnostic step than replacing adsorbent media on the assumption of normal aging.

Related Concepts / Tags

nitrogen-generation PSA pressure-swing-adsorption carbon-molecular-sieve cycle-valve bed-isolation product-purity oxygen-breakthrough cyclic-process

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.